ElectronicsForYou

Vol. 43 No. 2 FEBRUARY 2011

43rd Year of Publication

Since January 1969

www.efymag.com

ISSN-0013-516X

CONTENTS

32 **Technology Focus**
Ambient Electronics: Your Hidden Helper

44 **Solar Tips**
Extending Life of Your Solar SMF Battery

46 **Embedded**
Multicore Becomes Mainstream

54 **EFY Report**
Designing in India (Part 2): Independent Design Houses: One-Stop Shops for Global Clients

66 **Technology**
Can Graphene Electronics Take Over?

74 **EFY Report**
Key Component Distributors in India

98 **Innovation**
Video Surveillance System That Never Sleeps

102 **Test & Measurement**
Multimeters for High-End Testing

110 **Career**
At the Centre of Nuclear Energy

115 **Design**
Developing Graphics Library with DirectFB

117 **Construction**
Automatic Room Power Controller

120 **Construction**
Interfacing Nokia Colour LCD With AVR Microcontroller

129 **Software Section**
- Traffic Light Simulator
- Create OEM Logo Using Visual Basic.NET

159 **Interview**
"Unified communications is going to become a key competitive advantage"— Anil Kumar Jain, managing director, Siemens Enterprise Communications, India

Departments

EFY-CD 10
CD-THEME 12
WEBSITES 14
LETTERS 16
Q&As 18
TECHNOLOGY NEWS 22
CIRCUIT IDEAS 124
NEW PRODUCTS 133
INDUSTRY NEWS 136
BOOKSHELF 141
BIZ ADS 142
ELECTROONS 161
ELECTRONICS MART 162
SUBJECT & ORGANISATION INDEX 170
ADVERTISERS' INDEX 171

Style

84 **First Look!**
- New VAIO Laptops
- Ultra-wide 3D LED TV
- Smallest Colour Laser Printer
- Wireless Bluetooth Speakers
- Ultra-slim Scanner
- Remote Control to Laptop
- Windows 7 Smartphone
- Colourful Eee PCs

90 **Buyers' Guide**
Portable Media Players

Next Issue

TECHNOLOGY FOCUS
Robotics

BUYERS' GUIDE
Robot Kits

SURVEY/EFY REPORT
Test & Measurement

EDITOR : Ramesh Chopra

EDITORIAL CORRESPONDENCE : Editorial Secretary
Phone: 011-26810601
E-mail: editor.efy@efyindia.com
(Technical queries: efyhelp@efyindia.com)

SUBSCRIPTIONS & MISSING ISSUES : Customer Care Officer, EFY
Phone: 011-26810601 or 02 or 03
E-mail: support@efyindia.com

BACK ISSUES, BOOKS, CDs, PCBs etc. : Ritu Vij Sajwan,
D-100/5, Okhla Industrial Area, Phase 1
New Delhi 110020
Phone: 011-2637 1661, 26371662
E-mail: info@kitsnspares.com

EXCLUSIVE NEWSSTAND DISTRIBUTORS : IBH Books & Magazine Distributors Pvt Ltd
Asch No. 30, Below Mahalaxmi Bridge
Mahalaxmi, Mumbai-400034
Phone: 022-40487401, 40497402, 40497474, 40497413; Fax: 40497434
E-mail: circulations@ibhworld.com

ADVERTISEMENTS NEW DELHI (HEAD OFFICE) : Poonam Sharma
D-8/11, Okhla Industrial Area, Phase 1,
New Delhi 110020 INDIA
Phone: 011-26810601 or 02 or 03 Fax: 26817563
E-mail: efyemp@efyindia.com

MUMBAI : Flory D'Souza
207, Shivneeti Chambers, 49, Dr A.B. Road, Worli Naka
Mumbai 400018; Ph: 022-24950047, 24928520
Fax: 24954278; E-mail: efymum@efyindia.com

BENGALURU : Jayashree
No. 9, 17th Main, 1st Cross, HAL II Stage
Indiranagar, Bengaluru 560008
Ph: 080-25263956, 25260823; Fax: 25260394
E-mail: efyblore@efyindia.com

CHENNAI : DBS House, 31-A, Cathedral Garden Road
Near Palmgrove Hotel, Chennai 600034
Phone: 011-26810601 or 02 or 03 Fax: 011-26817563
E-mail: efymds@efyindia.com

KOLKATA : D.C. Mehta
61-A, Park Street, Kolkata 700016
Ph: 033-22296780, Telefax: 22650894
E-mail: efycal@efyindia.com; Mobile: 09432422552

PUNE : Sandeep Shandilya, Mobile: 09324099081
E-mail: efypune@efyindia.com

GUJARAT : Sunil Singh; Mobile: 09999661298
E-mail: efyahd@efyindia.com
Shambhushiva Patel; Mobile: 09825200451
Ph: 079-22139840; Telefax: 079-22139841

SINGAPORE : Ms Peggy Ding, Publicitas Singapore Pte Ltd
72, Bendemeer Road, #02-20,
The Luzerne, Singapore 339941
Phone: +65-6836 2272 Fax: +65-6297 7382
E-mail: pthuy@publicitas.com, singapore@publicitas.com

UNITED STATES : Ms Veronique Lemarque, E & Tech Media
Phone: +1 860 536 6677
E-mail: veronique@emargue@gmail.com

CHINA : Ms Terry Qin, Power Pioneer Group Inc.
3E Sorelle Building Bagua Forth Road, Futian Dis.
Shenzhen China, 518031
Phone: (86 755) 83729797; Fax: (86 21) 6455 2579
Mobile: (86) 13923802595, 18603515018
E-mail: terryqin@powerpioneergroup.com,
gaypterry@gmail.com

TAIWAN : Mr Jean Chen, J.K. Media
4F-7, No. 348, Sec. 2, Bade Rd., Songshan Dist.,
Taipei City 105, Taiwan
Phone: 886-2-87726780 ext 10; Fax: 886-2-87726787

Printed, published and owned by Ramesh Chopra. Printed at Nutech Photolithographers, D-30, Okhla Industrial Area, Phase-I, New Delhi, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase I, New Delhi 110020. Copyright 2011. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

Period	Newstand Price	You Pay	Digital	Overseas	
Year	(₹)	(₹)	(₹)	Digital	Print
Five	3,600	1,980	—	—	—
Three	2,160	1,400	—	—	—
One	720	540	360	US$ 10	US$ 100

For receiving both Summer and Winter editions of EFY Directories with Yellow Pages (each priced at ₹ 300), please add ₹ 800 for each year (including handling and forwarding).

Kindly add ₹ 50/- for outside Delhi cheques.

Please send payments only in favour of **EFY Enterprises Pvt Ltd**

Non-receipt of copies may be reported to support@efyindia.com mentioning your subscription number